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# OCNR: Stabilizing Self-Play by Mitigating Iteration-Collapse with One-Class Novelty Rewards

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Background

# Post-Training via Closed-Loop Self-Play

**Reinforcement Learning** improves LLM reasoning capabilities, in practice.

**Data Bottleneck:** Human-curated data is expensive, finite, and hard to scale.

SELF-PLAY CYCLE

**Questioner**
Creates novel questions

**Solver**
Generates answers

***Goal: open-ended self-improving loop achieving (near) zero-data continuous scaling.***

Background

# The Wall: Iteration Collapse

**Self-Play** pipelines offer a path toward open-ended improvement, in principle.

**Iteration Collapse:** Performance initially improves but subsequently degrades.

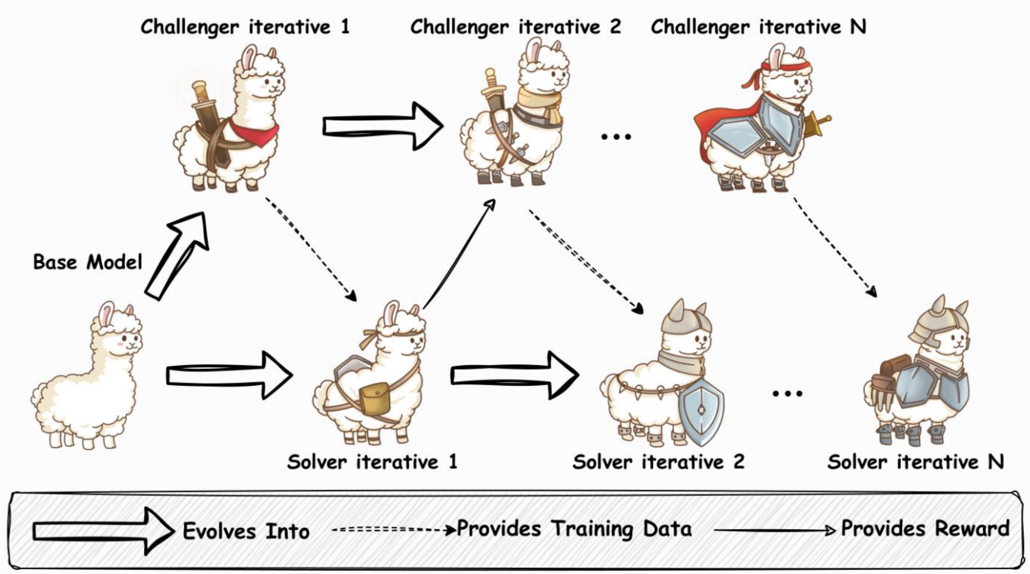


Image credit: R-Zero (*Huang et al., 2026*)

## Background

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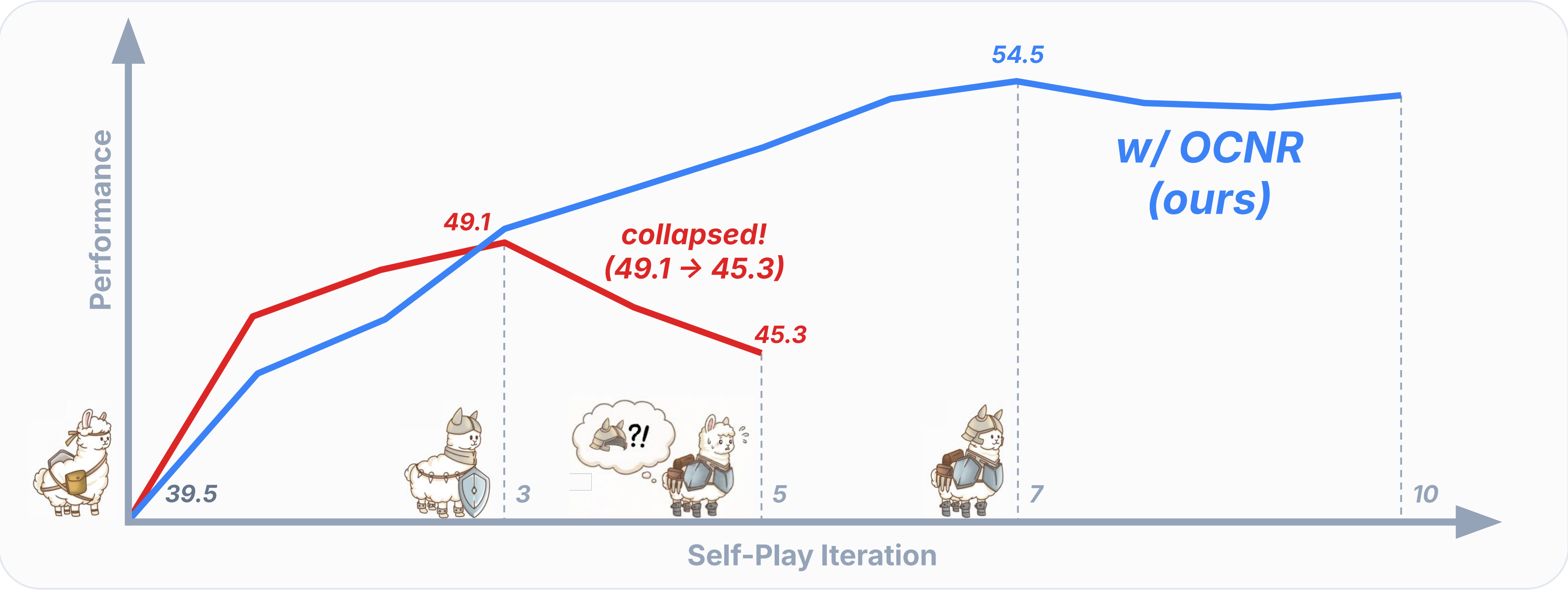


## Approach

# Novelty as the Key to Stability

CRITICAL

### (1) Validity

Questions should be solvable and well-formatted.

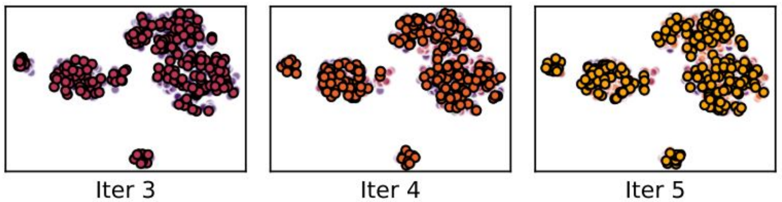
### (2) Difficulty

Questions should be tailored to the solver's level.

### (3) Diversity

Questions should encompass a wide range of types.

**Previous attempts for the diversity**
- Explicit prompting (Absolute Zero; *Zhao et al., 2025*).
- Batch-level BLEU penalties (R-Zero; *Huang et al., 2026*).




2D t-SNE of question embeddings across self-play iteration

***Problems are diverse within an iteration, but non-diverse across iterations.***

*The core issue is not just diversity but novelty: problems must be genuinely new from the solver's perspective.*

## Approach

# Novelty Quantification

### Seen Detector (via Learnable Prototype Vector)

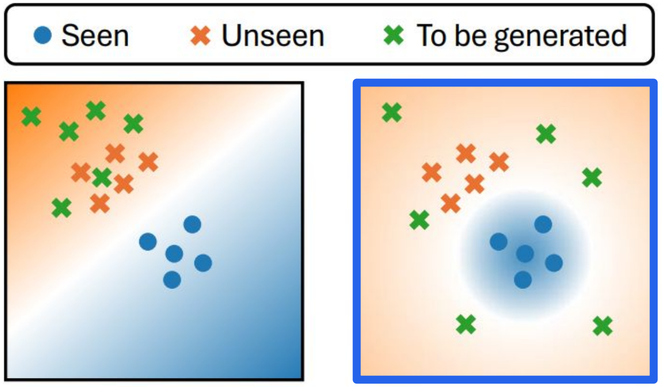
$$\mathrm{SEEN}(q_t) = \exp\left(-\frac{\|h(q_t) - w_s\|^2}{\tau}\right)$$

### One-Class Novelty Reward (OCNR)

$$R_Q^{\mathrm{nov}}(q_t) = 1 - \mathrm{SEEN}(q_t)$$

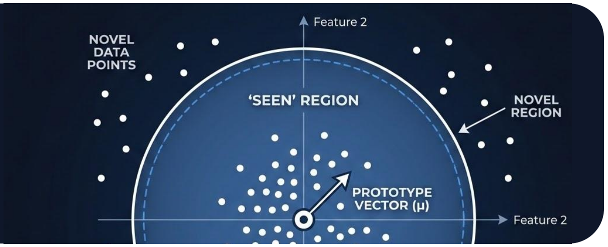
Conceptual illustration of novelty rewards

***Avoids global half-space penalties of sigmoid boundaries!***




(a) Sigmoid-based (b) Distance-based

***OCNR penalizes redundant tasks while targeting the solver's actual blind spots.***



Approach

# The Alternating Self-Play Loop

**01**

### Generate & Split Questions

Construct a balanced **Seen/Unseen** sets.

**02**

### Update Solver
(baseline reward)

Train the solver using only the **Seen** subset.

**03**

### Update Detector
(binary cross-entropy)

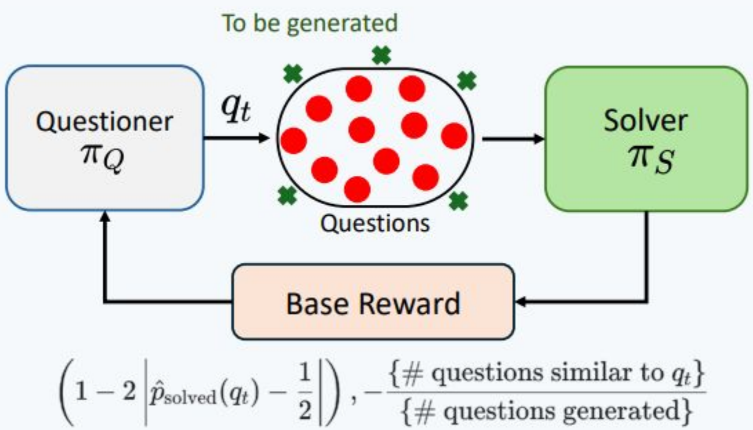
Train the detector on **Seen/Unseen** sets.

**04**

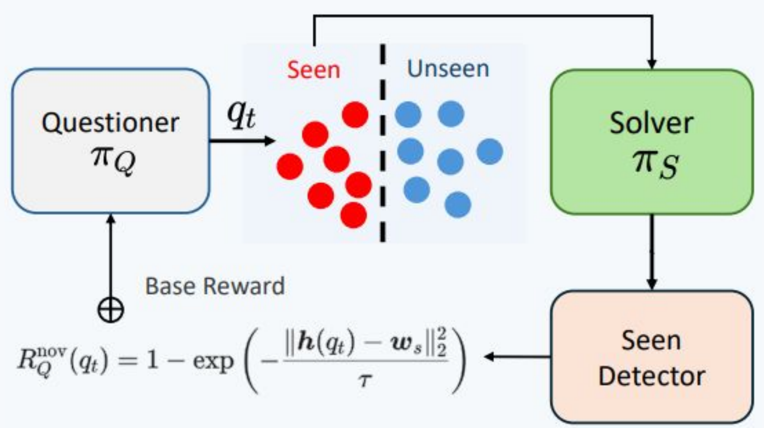
### Update Questioner
(baseline + novelty reward)

Update the Questioner by adding the OCNR novelty term to its base reward.

### *Original Pipeline (R-Zero)*




### *OCNR Pipeline (R-Zero w/ OCNR)*



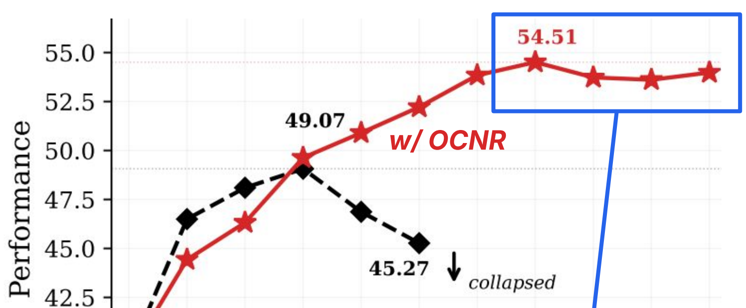
## Main Results

# Mathematical Reasoning

| Method | Iter 1 | Iter 2 | Iter 3 | Iter 4 | Iter 5 |
|---|---|---|---|---|---|
| R-Zero | 46.5 | 48.1 | **49.1** | **46.9** | **45.3** |
| w/ OCNR (Ours) | 44.4 | 46.3 | 49.6 | 50.9 | **52.2** |

### 10-Iteration Extension

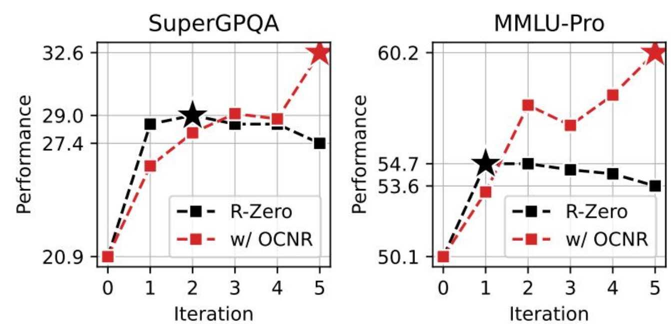
Extended testing demonstrates that OCNR sustains improvement up to **Iteration 7**.




Notably, it maintains **the stable plateau** through Iteration 10, preventing post-peak degradation.

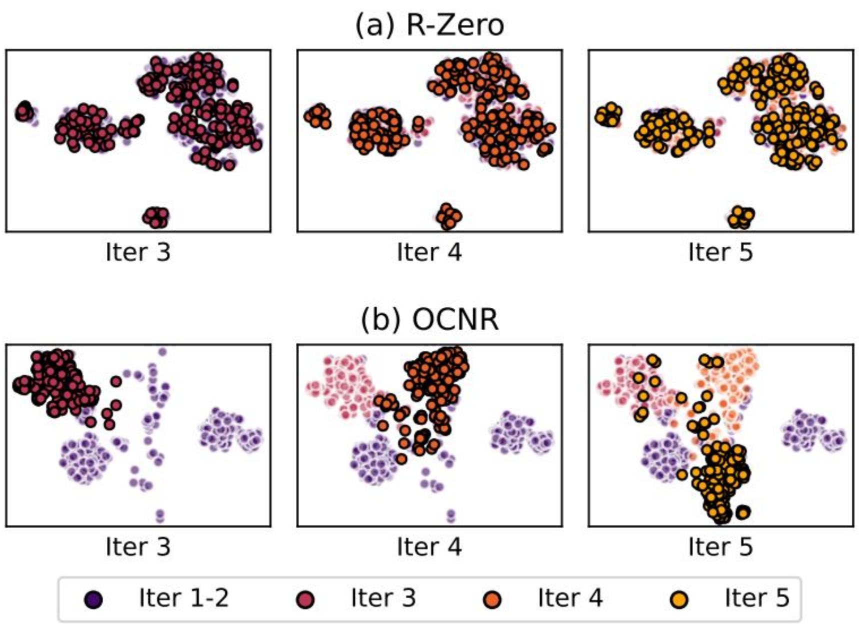
### Cross-Domain Leap

Math-centric self-play successfully transfers to **general-purpose reasoning** datasets at test time.




Notably, it enhances the self-play pipeline itself, moving **beyond task-specific gains**.

# Diversity and Novelty of Questions



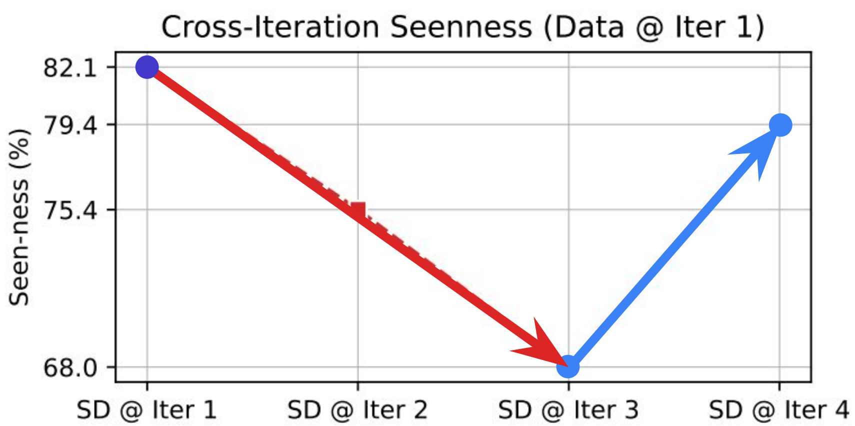

2D t-SNE of question embeddings across self-play iteration

***Non-diverse across iterations!***

***Diverse across iterations!***

***Our strategy yields not only diverse but specifically novel questions.***

# Re-emergence of Forgotten Questions

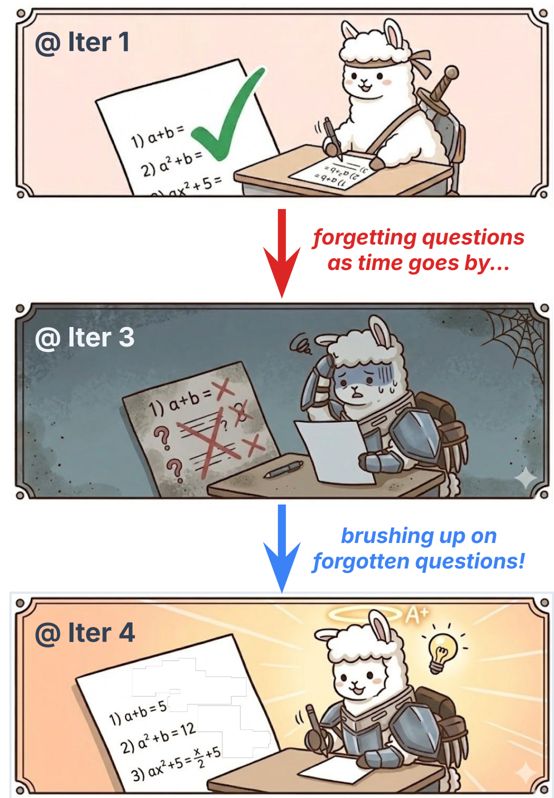



**Question**

Let $a, b, c$ be the roots of the cubic polynomial $x^3 - 5x^2 + 6x - 1 = 0$.
Find the value of the expression $\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$.

**Re-emerged**

Let $x_1, x_2, x_3$, and $x_4$ denote the roots of the equation: $x^4 - 6x^3 + 5x^2 - 4x + 2 = 0$
Determine the value of the sum: $\sum_{k=1}^{4} \frac{1}{x_k}$

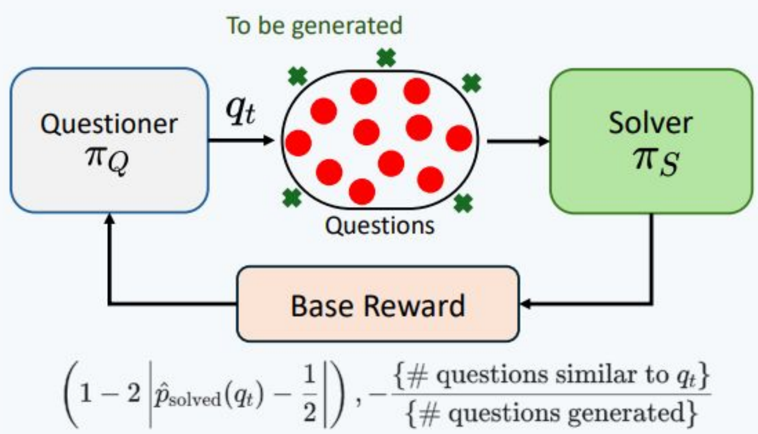


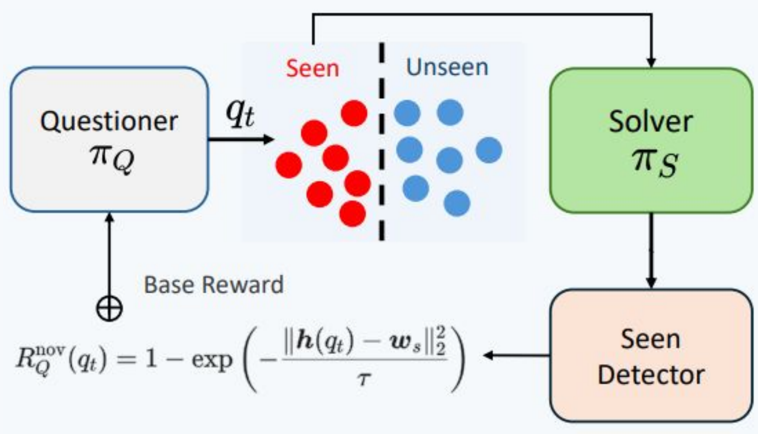

**_Our strategy makes a questioner re-discovers forgotten concepts._**

## Analytical Results

# Over-generation of Questions

| Method | Iter 1 | Iter 2 | Iter 3 | Iter 4 | Iter 5 |
|---|---|---|---|---|---|
| R-Zero | 46.5 | 48.1 | **49.1** | **46.9** | **45.3** |
| R-Zero (2x) | 47.5 | **49.2** | **47.8** | **46.1** | **44.0** |
| w/ OCNR (Ours) | 44.4 | 46.3 | 49.6 | 50.9 | **52.2** |




*Our strategy goes beyond the simple benefits of over-generation.*

# Summary and Open Directions

- **Core Finding:** iteration collapse is fundamentally caused by cross-iteration distribution degeneration into familiar spaces.
- **Proven Solution:** OCNR stops redundancy at the source, offering a simple, computational-efficient plug-in for data-free scaling.
- **Future Extensions:** moving beyond our memory-based single prototype approach; improving sample efficiency; integrating safety guardrails.

**Thank you!**